CHAPTER 2

STRUCTURE OF ATOM

CHAPTER ESSENCE AT A GLANCE:
Explores quantum mechanical atomic architecture, wave-particle duality, orbital quantization, and electronic configuration principles.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Dual Nature of Matter:** Microscopic particles exhibit dual wave-particle character with de Broglie wavelength inversely proportional to momentum.

• **Heisenberg Uncertainty Principle:** Position and momentum of subatomic particles cannot be measured simultaneously with arbitrary precision.

• **Quantum Numbers & Orbitals:** Four quantum numbers (n, l, m_l, m_s) describe orbital size, angular shape, orientation, and electron spin.

• **Aufbau & Hund Principles:** Orbitals fill in order of increasing (n + l) energy; pairing occurs only after degenerate orbitals receive one electron each.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

de Broglie Wavelength of Moving Matter

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Heisenberg Uncertainty Principle

$$\bar{\nu} = \frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Rydberg Formula for Hydrogen Spectral Series

$$E_n = -\frac{2.18 \times 10^{-18}}{n^2}\ \text{J} = -\frac{13.6}{n^2}\ \text{eV}$$

Quantized Energy of nth Bohr Orbit

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Chromium (3d5 4s1) and Copper (3d10 4s1) have anomalous configurations due to extra stability of half-filled and full subshells.

• **Alert:** De Broglie wavelengths for macroscopic objects are too minuscule (order of 10^{-34} m) to exhibit observable wave diffraction.

• **Alert:** In Rydberg equation calculations, n2 must always be strictly greater than n1 to produce positive wavenumber.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State Pauli's Exclusion Principle and Hund's Rule of Maximum Multiplicity. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Pauli's Exclusion Principle:** No two electrons in an isolated atom can have the identical set of all four quantum numbers; an orbital can hold a maximum of two electrons with anti-parallel spins. **[1 Mark]**

• **Hund's Rule of Maximum Multiplicity:** Electron pairing in degenerate orbitals (p, d, f) does not occur until each orbital in the subshell is singly occupied with parallel spins. **[1 Mark]**

• **Key Exam Result:** Governs ground state electron configurations and ensures minimum inter-electronic repulsion.

2. State de Broglie relation. Why is the wave nature of macroscopic bodies not detectable in daily life? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Statement:** All moving material particles possess wave characteristics with wavelength inversely proportional to momentum: lambda equals h divided by momentum p. **[1 Mark]**

• **Macroscopic Justification:** Because Planck's constant h is extremely small (6.626×10^{-34} J s), macroscopic bodies with large masses possess de Broglie wavelengths of order 10^{-34} m, far below observable physical detection. **[1 Mark]**

• **Key Exam Result:** Wave nature is significant only for subatomic particles of sub-microscopic mass.

3. Write the Rydberg formula for hydrogen emission spectrum. List the five spectral series of hydrogen with their respective spectral regions. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Rydberg Equation:** Wavenumber equals R_H multiplied by (1 divided by n1 squared minus 1 divided by n2 squared), where R_H = 109677 per cm and n2 > n1. **[1 Mark]**

• **Lyman and Balmer Series:** Lyman series (n1 = 1, n2 = 2, 3...) lies in Ultraviolet region. Balmer series (n1 = 2, n2 = 3, 4...) lies in Visible spectrum. **[1 Mark]**

• **Infrared Series:** Paschen (n1 = 3), Brackett (n1 = 4), and Pfund (n1 = 5) series all lie in the Infrared region of the electromagnetic spectrum. **[1 Mark]**

• **Key Exam Result:** Balmer series is the only spectral series of hydrogen falling in the visible wavelength range.

$$\bar{\nu} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \quad (\text{Balmer: } n_1 = 2,\ \text{Visible})$$

Hydrogen Spectral Series Formulation

4. Explain the anomalous electronic configurations of Chromium (Z=24) and Copper (Z=29) based on exchange energy and symmetrical distribution. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Expected vs Actual Configurations:** Expected Cr is [Ar] 3d4 4s2, but actual is [Ar] 3d5 4s1. Expected Cu is [Ar] 3d9 4s2, but actual is [Ar] 3d10 4s1. **[1 Mark]**

• **Symmetrical Electron Distribution:** Completely filled subshells (d10) and exactly half-filled subshells (d5) possess spherical symmetry and uniform electron shielding, lowering potential energy. **[1 Mark]**

• **Maximum Exchange Energy:** Electrons with parallel spins in degenerate orbitals can exchange positions; maximum exchange interactions occur in half-filled and fully-filled configurations, releasing maximum stabilizing exchange energy. **[1 Mark]**

• **Key Exam Result:** Extra stability of d5 and d10 subshells drives 4s electron promotion to 3d.

5. A microscopic particle of mass 0.1 kg is moving with a velocity of 10 m/s with an uncertainty of 1%. Calculate the minimum uncertainty in its position (h = 6.626 × 10^{-34} J s). **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Step 1 (Velocity Uncertainty):** Δv = 1% of 10 m/s = (1 divided by 100) × 10 = 0.1 m/s. **[1 Mark]**

• **Step 2 (Momentum Uncertainty):** Δp = m × Δv = 0.1 kg × 0.1 m/s = 0.01 kg m/s = 10^{-2} kg m/s. **[1 Mark]**

• **Step 3 (Position Uncertainty Calculation):** Δx = h divided by (4 π Δp) = (6.626 × 10^{-34}) divided by (4 × 3.1416 × 0.01) = 5.27 × 10^{-33} m. **[1 Mark]**

• **Key Exam Result:** Uncertainty in position = 5.27 × 10^{-33} m.

$$\Delta x \geq \frac{h}{4\pi m \Delta v} = \frac{6.626 \times 10^{-34}}{4\pi \times 0.1 \times 0.1} = 5.27 \times 10^{-33} \text{ m}$$

Position Uncertainty Evaluation

6. Define the four quantum numbers and state the physical significance of each. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Principal Quantum Number (n):** Designates main energy shell (n = 1, 2, 3...) and determines overall orbital size and distance from the nucleus. **[0.5 Mark]**

• **Azimuthal (Orbital Angular Momentum) Number (l):** Determines orbital three-dimensional geometry and shape (l = 0 for s, 1 for p, 2 for d, 3 for f). **[1 Mark]**

• **Magnetic Quantum Number (m_l):** Dictates spatial orientation of orbitals in magnetic field (values from -l to +l, total 2l + 1 orientations). **[1 Mark]**

• **Spin Quantum Number (m_s):** Represents intrinsic spin angular momentum of electron (+0.5 or -0.5). **[0.5 Mark]**

• **Key Exam Result:** A complete set of four quantum numbers uniquely defines the quantum state of any atomic electron.

7. Calculate the wavelength and frequency of the radiation emitted when an electron in a hydrogen atom undergoes transition from n=4 to n=2. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Step 1 (Wavenumber Calculation):** Wavenumber = 109677 × (0.25 - 0.0625) = 109677 × 0.1875 = 20564.4 per cm. **[1 Mark]**

• **Step 2 (Wavelength Evaluation):** λ = 1 divided by 20564.4 per cm = 4.86 × 10^{-5} cm = 486 nm (cyan line in visible Balmer series). **[1 Mark]**

• **Step 3 (Frequency Calculation):** Frequency ν = c divided by λ = (3 × 10^{8} m/s) divided by (4.86 × 10^{-7} m) = 6.17 × 10^{14} Hz. **[1 Mark]**

• **Key Exam Result:** Wavelength = 486 nm; Frequency = 6.17 × 10^{14} Hz (Balmer H-β line).

$$\frac{1}{\lambda} = 109677\left(\frac{1}{4} - \frac{1}{16}\right) \rightarrow \lambda = 486 \text{ nm}, \ \nu = 6.17 \times 10^{14} \text{ Hz}$$

Balmer Transition Parameters